

Wenda Zhang

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Education

Doctor of Philosophy: Marine Science

Stony Brook University

Advisor: Prof. Christopher L. P. Wolfe

2017 - 2022

Stony Brook, NY, U.S.

Bachelor of Science: Marine Science

Ocean University of China

2013 - 2017

Qingdao, China

Research Interests

Ocean mesoscale eddies, and their impact on mixing, large-scale ocean circulations and climate; parameterization of oceanic mesoscale and submesoscale processes; geophysical fluid dynamics; development of idealized and realistic ocean models.

Appointments

Postdoctoral Research Associate

Host: Dr. Stephen M. Griffies

August 2022-present

Princeton University

- Studying the vertical structure of mesoscale and submesoscale ocean eddies in idealized models, and developing parameterizations of this structure for use in global climate and prediction models.
- Implementing and improving a weighted essentially non-oscillatory (WENO) advection scheme and the kinetic energy backscatter parameterization in GFDL Modular Ocean Model (MOM6).

Research Assistant

Advisor: Prof. Christopher L. P. Wolfe

2018-2022

Stony Brook University

- Quantified potential vorticity (PV) transport due to coherent eddies and identified the relation between the PV diffusivity and dispersion of coherent eddies in a two-layer quasigeostrophic model.
- Analyzed the vertical structure of the mesoscale tracer diffusivity in an idealized ocean configuration of the MITgcm.

Undergraduate Researcher

Supervisor: Prof. Xueen Chen

2016 - 2017

Ocean University of China

- Diagnosed the energy transfer between mesoscale eddies and mean flows through barotropic and baroclinic pathways in the South China Sea based on the Global Hybrid Coordinate Ocean Model (HYCOM) hindcasts.

Publications

In Progress

- **Zhang, W.**, Y. Kuo, S. Silvestri, A. Adcroft, R.W. Hallberg, S.M. Griffies, 2025: A WENO finite-volume scheme for the evolution of potential vorticity in isopycnal ocean models. *Submitted to Journal of Advances in Modeling Earth Systems*. Preprint DOI: [10.22541/essoar.175380391.18723979/v1](https://doi.org/10.22541/essoar.175380391.18723979/v1).
- Wang, S., **Zhang W.**, Kido S., Sasaki H., Qiu B., Guo X., 2025: Understanding energy cascade in the Northwest Pacific using a submesoscale permitting OGCM. *Submitted to SCIENCE CHINA Earth Sciences*.
- Feng, Z., Zhang Z., Zhang J., **Zhang W.**, Yuan M., Jing Z., Zhao W., Tian J., 2025: Implementation of a new parameterization of submesoscale vertical flux in an eddy-resolving model in the North Pacific. *Under revision on Ocean Modelling*.
- Pudig, M., **Zhang W.**, Smith K.S., Zanna L., 2025: Parameterizing isopycnal mixing via kinetic energy backscatter in an eddy-permitting ocean model. *Submitted to Journal of Advances in Modeling Earth Systems*. Preprint DOI: [10.22541/essoar.175805328.85239601/v1](https://doi.org/10.22541/essoar.175805328.85239601/v1).
- **Zhang, W.**, A. Adcroft, E. Yankovsky, S.M. Griffies, R.W. Hallberg, 2024: A scale-dependent vertical structure for mesoscale energy backscatter parameterizations. *In preparation for Journal of Advances in Modeling Earth Systems*

Peer Reviewed

- Zhang, Z., Chang J., Zhang X., and **Zhang W.**, 2025: Mixed transitional layer instability: A mechanism for deep-penetrating submesoscale processes in the subtropical upper ocean. *Journal of Physical Oceanography*
- Griffies, S. M., Adcroft, A., Beadling, R. L., Bushuk, M., Chang, C. Y., Drake, H. F., ..., **Zhang, W.**, Zhao, M. (2024). The GFDL-CM4X climate model hierarchy, Part I: model description and thermal properties. *Journal of Advances in Modeling Earth Systems*

- Griffies, S. M., Adcroft, A., Beadling, R. L., Bushuk, M., Chang, C. Y., Drake, H. F., ..., **Zhang, W.**, Zhao, M. (2024). The GFDL-CM4X climate model hierarchy, Part II: case studies. *Journal of Advances in Modeling Earth Systems*
- Lobo, M., Griffies, S.M. and **Zhang, W.**, 2024. Vertical structure of baroclinic instability in a three-layer quasi-geostrophic model over a sloping bottom. *Journal of Physical Oceanography*, DOI: <https://doi.org/10.1175/JPO-D-23-0203.1>
- **Zhang, W.**, S.M. Griffies, R.W. Hallberg, Y. Kuo, and C.L.P. Wolfe, 2024: The role of surface potential vorticity in the vertical structure of mesoscale eddies in wind-driven ocean circulations. *Journal of Physical Oceanography*, DOI: <https://doi.org/10.1175/JPO-D-23-0203.1>
- **Zhang, W.**, C.L.P. Wolfe, 2024: Inferring tracer diffusivity from coherent mesoscale eddies. *Journal of Advances in Modeling Earth Systems*, 16, e2023MS004004. <https://doi.org/10.1029/2023MS004004>
- **Zhang, W.**, C.L.P. Wolfe, 2022: On the vertical structure of oceanic mesoscale tracer diffusivities. *Journal of Advances in Modeling Earth Systems*, 14, e2021MS002891. <https://doi.org/10.1029/2021MS002891>
- **Zhang, W.**, C.L.P. Wolfe, R. Abernathey, 2020: Role of surface-layer coherent eddies in potential vorticity transport in quasi-geostrophic turbulence driven by eastward shear. *Fluids*, 5(1), p.2, doi: [10.3390/fluids5010002](https://doi.org/10.3390/fluids5010002)

Invited Presentations

“Scale- and flow-aware subgrid closure for mesoscale eddies in ocean climate models”

University of Tokyo and Japan Agency for Marine-Earth Science and Technology (JAMSTEC)

January 2025

“The role of surface potential vorticity in the vertical structure of mesoscale eddies”

NOAA Coastal Ocean Modeling Seminars, online

July 2024

Conference Presentations

- “An implicit representation of subgrid potential vorticity mixing in isopycnal ocean models”, Climate Process Team: Ocean Transport and Eddy Energy Annual Meeting, New York City, NY, August 2025 (**oral presentation**)
- “Sensitivity of the kinetic energy backscatter parameterization to its vertical structure in eddy-permitting ocean simulations”, CLIVAR Ocean Model Development Panel and COMMODORE Workshop, Boulder, CO, September 2024 (**poster presentation**)
- “Sensitivity of the kinetic energy backscatter parameterization to its vertical structure in eddy-permitting ocean simulations”, Climate Process Team: Ocean Transport and Eddy Energy Annual Meeting, Providence, RI, August 2024 (**oral presentation**)
- “The role of surface potential vorticity in the vertical structure of mesoscale eddies in wind-driven ocean circulations”, 24th Conference on Atmospheric and Oceanic Fluid Dynamics, Burlington, VT, June 2024 (**oral presentation**)
- “A scale-dependent vertical structure for energy backscatter parameterizations”, EGU conference, Vienna, Austria, April 2024 (**oral presentation**)
- “The role of surface potential vorticity in the vertical structure of mesoscale eddies”, Ocean Sciences Meeting, New Orleans, LA, February 2024 (**oral presentation**)
- “A scale-dependent vertical structure for mesoscale energy backscatter parameterizations”, CESM Ocean Model Working Group Meeting, Virtual, February 2024 (**oral presentation**)
- “Scale-dependent vertical structure of eddy kinetic energy in an adiabatic ocean model”, Climate Process Team: Ocean Transport and Eddy Energy Annual Meeting, Woods Hole, MA, May 2023 (**oral presentation**)
- “Scale-dependent vertical structure of eddy kinetic energy in an idealized isopycnal ocean model”, CESM Ocean Model Working Group Meeting, Virtual, February 2023 (**oral presentation**)
- “Inferring tracer diffusivity from coherent mesoscale eddies”, 23rd Conference on Atmospheric and Oceanic Fluid Dynamics, Breckenridge, CO, June 2022 (**poster**)
- “On the vertical structure of oceanic mesoscale tracer diffusivities”, Climate Process Team: Ocean Transport and Eddy Energy Annual Meeting, Boulder, CO, April 2022 (**oral presentation**)
- “What determines the vertical structure of mesoscale tracer diffusivity?”, Ocean Sciences Meeting, Virtual, March 2022 (**oral presentation**)
- “Vertical structure of tracer diffusivity in an idealized basin circulation model”, CESM Ocean Model Working Group Meeting, Virtual, February 2021 (**oral presentation**)
- “Diffusive versus nondiffusive properties of coherent ocean eddies”, Ocean Sciences Meeting, San Diego, CA, February 2020 (**eLightning presentation**)
- “Role of coherent eddies in potential vorticity transport in two-layer quasigeostrophic turbulence”, 22nd Conference on Atmospheric and Oceanic Fluid Dynamics, Portland, ME, June 2019 (**oral presentation**)

Professional Services

Conference Session Primary Chair 2024 Ocean Sciences Meeting, Session PL41A & PL53A: Multiscale Eddy Dynamics and Tracer Transport: Bridging Observations, Theory, and Modeling	<i>New Orleans, LA</i> February 2024
Conference Session Co-Chair 2025 Xiamen Symposium on Marine Environmental Sciences, Session 35: Eddy variability in the ocean and atmosphere: dynamics, parameterization and prediction	<i>Xiamen, China</i> January 2025
Journal Reviewer Reviewer for Journal of Physical Oceanography, Journal of Advances in Modeling Earth Systems, Journal of Geophysical Research - Oceans, Climate Dynamics	2022 - Present

Teaching Experience

Teaching Assistant Oceanography (MAR 104) Physics for Environmental Studies (ENS 119)	<i>Stony Brook University</i> Fall 2017 and Fall 2018 Spring 2018 and Spring 2021
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Awards and Honors

2020	Maze-Landean Travel Award , Stony Brook University
2019	Distinguished Travel Award , Stony Brook University
2019	IACS Travel Award , Stony Brook University
2016	The Outstanding Student Award , Ocean University of China
2016	The Scholarship Award for Participation in Social Activities , Ocean University of China
2015	The First-Class Scholarship Award for Excellence in Academic Work , Ocean University of China
2015	Second Prize in Physics Competition of Chinese College Students (non-physics major) , Chinese Physical Society
2014	First Prize in Mathematics Competition of Chinese College Students (non-mathematics major) , Chinese Mathematical Society

Volunteer Services

NJ Ocean Fun Days Volunteer	<i>Island Beach State Park</i> May 2023
Workshop for Boys and Girls Club Volunteer	<i>Mercer County, NJ</i> March 2023
Qingdao Red Cross Society Volunteer	<i>Qingdao, China</i> June-July 2013

Programming Skills

Programming	Python, Matlab
Computer Language	Fortran